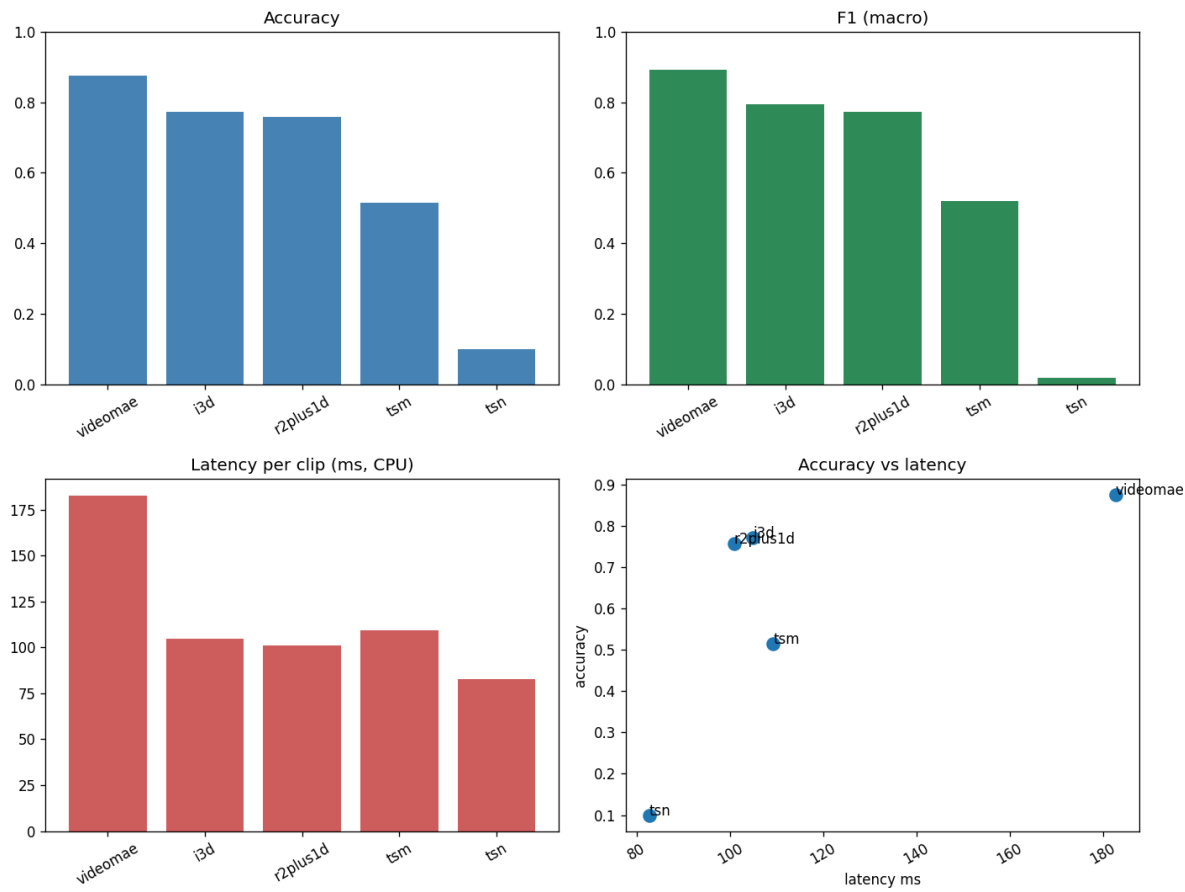


# Action Recognition — Model Comparison Report

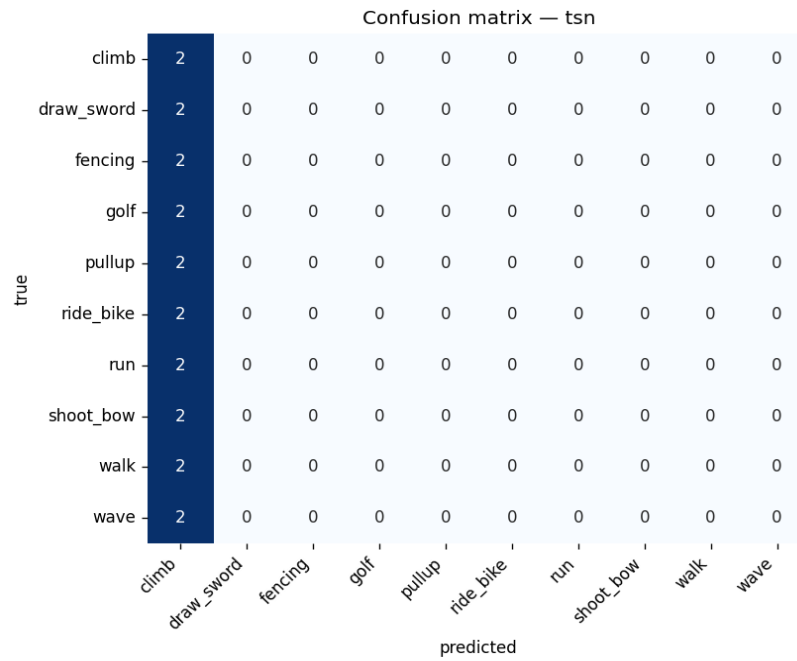
Dataset: HMDB51 subset (10 classes). Classes: climb, draw\_sword, fencing, golf, pullup, ride\_bike, run, shoot\_bow, walk, wave.

| model    | accuracy | precision_macro | recall_macro | f1_macro | latency_s_per_clip | fps     | stability | mean_confidence | params   | size_mb |
|----------|----------|-----------------|--------------|----------|--------------------|---------|-----------|-----------------|----------|---------|
| videomae | 0.876    | 0.894           | 0.892        | 0.892    | 0.183              | 87.543  | 1.0       | 0.664           | 86234890 | 328.96  |
| i3d      | 0.772    | 0.825           | 0.781        | 0.795    | 0.105              | 152.484 | 0.95      | 0.476           | 27244362 | 104.132 |
| r2plus1d | 0.757    | 0.79            | 0.772        | 0.772    | 0.101              | 158.603 | 0.85      | 0.443           | 31305255 | 119.517 |
| tsm      | 0.514    | 0.554           | 0.52         | 0.521    | 0.109              | 73.203  | 0.862     | 0.265           | 11181642 | 42.691  |
| tsn      | 0.1      | 0.01            | 0.1          | 0.018    | 0.083              | 96.805  | 1.0       | 0.178           | 11181642 | 42.691  |

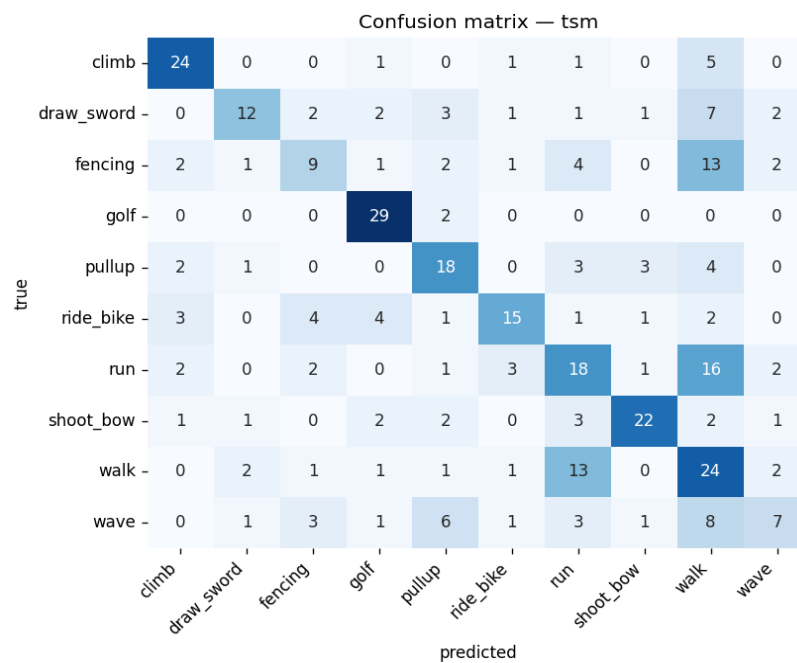
## Comparison charts



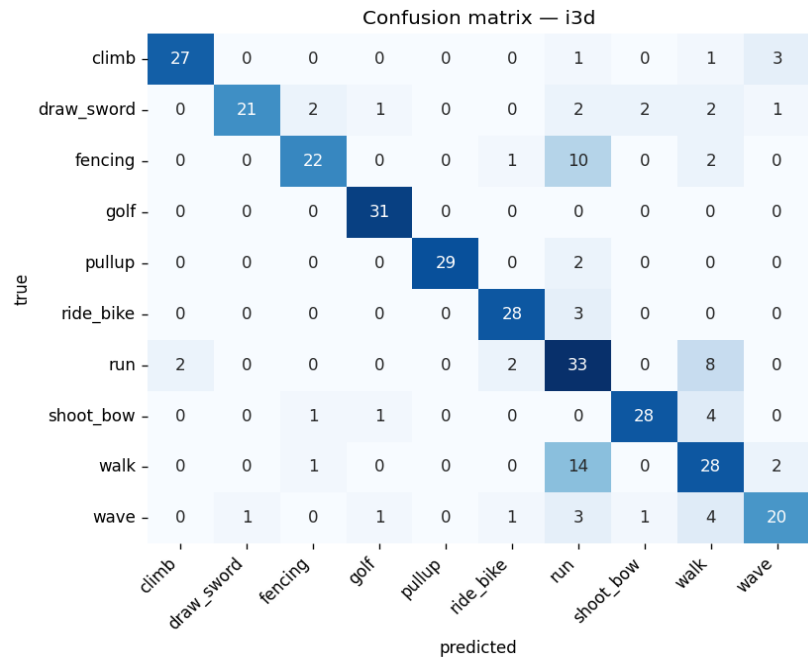
## Confusion matrix — tsn



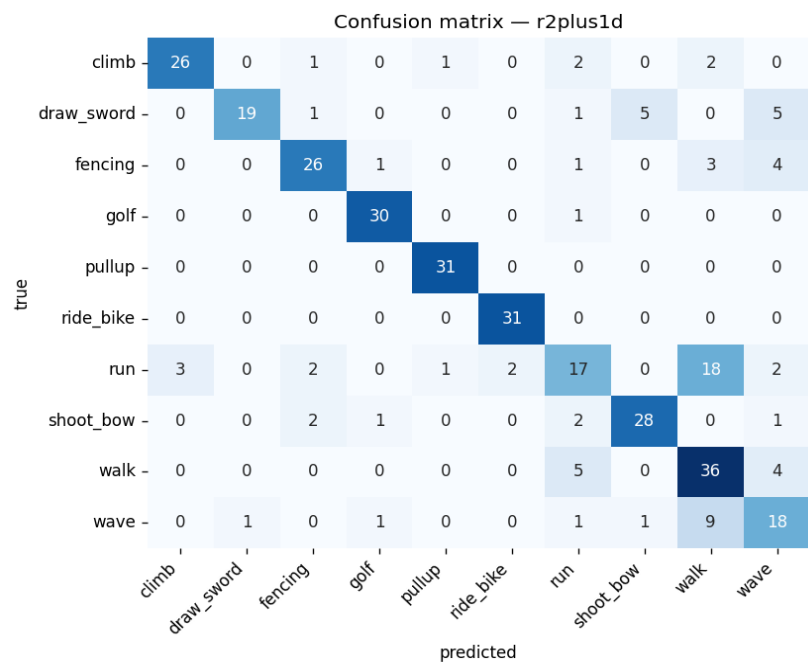
### Confusion matrix — tsm



### Confusion matrix — i3d



### Confusion matrix — r2plus1d



### Confusion matrix — videomae

|            | climb | draw_sword | fencing | golf | pullup | ride_bike | run | shoot_bow | walk | wave |
|------------|-------|------------|---------|------|--------|-----------|-----|-----------|------|------|
| climb      | 31    | 0          | 0       | 0    | 0      | 0         | 1   | 0         | 0    | 0    |
| draw_sword | 0     | 29         | 1       | 0    | 0      | 0         | 0   | 0         | 0    | 1    |
| fencing    | 0     | 0          | 33      | 0    | 0      | 0         | 1   | 0         | 1    | 0    |
| golf       | 0     | 0          | 0       | 31   | 0      | 0         | 0   | 0         | 0    | 0    |
| pullup     | 0     | 0          | 0       | 0    | 31     | 0         | 0   | 0         | 0    | 0    |
| ride_bike  | 0     | 0          | 0       | 0    | 0      | 30        | 0   | 0         | 1    | 0    |
| run        | 0     | 0          | 3       | 0    | 0      | 0         | 33  | 0         | 8    | 1    |
| shoot_bow  | 0     | 0          | 0       | 1    | 0      | 0         | 0   | 31        | 1    | 1    |
| walk       | 0     | 0          | 0       | 0    | 0      | 0         | 12  | 0         | 28   | 5    |
| wave       | 1     | 0          | 1       | 0    | 0      | 0         | 0   | 0         | 3    | 26   |

|            | climb | draw_sword | fencing | golf | pullup | ride_bike | run | shoot_bow | walk | wave |
|------------|-------|------------|---------|------|--------|-----------|-----|-----------|------|------|
| climb      | 31    | 0          | 0       | 0    | 0      | 0         | 1   | 0         | 0    | 0    |
| draw_sword | 0     | 29         | 1       | 0    | 0      | 0         | 0   | 0         | 0    | 1    |
| fencing    | 0     | 0          | 33      | 0    | 0      | 0         | 1   | 0         | 1    | 0    |
| golf       | 0     | 0          | 0       | 31   | 0      | 0         | 0   | 0         | 0    | 0    |
| pullup     | 0     | 0          | 0       | 0    | 31     | 0         | 0   | 0         | 0    | 0    |
| ride_bike  | 0     | 0          | 0       | 0    | 0      | 30        | 0   | 0         | 1    | 0    |
| run        | 0     | 0          | 3       | 0    | 0      | 0         | 33  | 0         | 8    | 1    |
| shoot_bow  | 0     | 0          | 0       | 1    | 0      | 0         | 0   | 31        | 1    | 1    |
| walk       | 0     | 0          | 0       | 0    | 0      | 0         | 12  | 0         | 28   | 5    |
| wave       | 1     | 0          | 1       | 0    | 0      | 0         | 0   | 0         | 3    | 26   |